ODE Definitions and Terminology

- An **IVP** is an **ODE** relating $\boldsymbol{y}(t)$ to its derivatives, with an **IC** $\boldsymbol{y}(t_0) = y_0$
  - Highest derivative is **Order**. A **Solution** satisfies ODE and IC on $a < t_0 < b$
  - **Autonomous** no explicit t. **Equilibrium** a constant solution $\boldsymbol{f}(\boldsymbol{y}_*) = \boldsymbol{0}$
  - An ODE **General Solution** contains all solutions with constants determined by ICs.
- **Separable** form $y' = g(t)h(y)$; linear **Standard Form** is $y' + p(t)y = q(t)$
- $\boldsymbol{y}' = \boldsymbol{A}(t)\boldsymbol{y} + \boldsymbol{b}(t)$ is **Linear**.
  $\boldsymbol{A}(t)$ constant is **Constant Coefficient (CC)**. $b = 0$ is **Homogeneous**.
- The general solution of CC homogeneous ODE $\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y}$ is

$$\boldsymbol{y}(t) = \sum_{i=1}^{n} c_i e^{\lambda_i t} \boldsymbol{v}_i$$

  provided $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots \boldsymbol{v}_n$ is a complete LI set of eigenvectors for $\boldsymbol{A}$.
  The constants $\boldsymbol{c}$ satisfy $\boldsymbol{V}\boldsymbol{c} = \boldsymbol{y}(0)$.